North Ridge Copper Mine Expansion & Processing Capacity Program

Mine Development, Processing Plant Upgrade, Contractor Records, Production Data & Financial Information

Synthetic raw non-IT project document for testing the AI Project Intelligence & Risk Advisor. The document contains project evidence only. It deliberately does not state a risk classification, risk probability, health score, blocker assessment, future prediction or recommendation.

1. Document Control

Field	Value
Organization	Redwood Mining Cooperative
Project ID	RMC-NRCP-2025-019
Industry	Mining / Metals / Heavy Industrial Operations
Project Type	Mine Expansion and Processing Plant Upgrade
Project Manager	David Morgan
Executive Sponsor	Helen Foster — Chief Operating Officer
Site General Manager	Robert Chen
Engineering Manager	Laura Bennett
Operations Manager	Carlos Ramirez
Finance Manager	Nina Shah
Start Date	15 March 2025
Original Completion	30 September 2026
Reporting Date	19 August 2026
Current Forecast Completion	18 March 2027
Current Phase	Construction, Equipment Installation and Commissioning
Reporting Frequency	Daily site coordination; weekly project review; monthly steering review

2. Organization and Project Background

Redwood Mining Cooperative operates open-pit copper and associated mineral-processing assets in a remote highland region. The North Ridge operation supplies concentrate to regional smelters and export customers.

The North Ridge Expansion Program was initiated to increase ore-processing capacity, extend supporting infrastructure and improve material-handling capacity. The program includes pit-access works, a primary crusher upgrade, conveyor modifications, additional flotation equipment, tailings-pump upgrades, water infrastructure, electrical works and a new maintenance workshop.

The project is being executed while the existing operation remains active. Construction crews, mining operations, maintenance teams and contractors must coordinate access to shared roads, power infrastructure and processing areas.

3. Business Objectives

Increase processing capacity from approximately 22,000 tonnes per day to 31,000 tonnes per day.

Increase ore-handling capacity at the primary crusher and conveyor system.

Improve plant availability through additional pumping and electrical capacity.

Expand water-recovery and process-water infrastructure.

Create additional maintenance capacity for the expanded operation.

Complete construction and commissioning before the next major production cycle.

Maintain safe operation of the existing mine and processing plant throughout the project.

4. Scope and Exclusions

Included: crusher upgrade, conveyor extension, flotation expansion, pump-station upgrade, water infrastructure, electrical substation work, maintenance workshop, haul-road modifications, contractor facilities, commissioning and operator training.

Excluded: new mining fleet procurement, development of a second open pit, construction of a new smelter, long-term rail infrastructure and expansion of the company's external logistics network.

5. Governance and Stakeholders

Role	Person / Group	Responsibility	Cadence
Executive Sponsor	Helen Foster	Funding and decisions	Monthly
Project Manager	David Morgan	Schedule/cost/coordination	Daily/weekly
Site General Manager	Robert Chen	Mine operations	Daily
Engineering Manager	Laura Bennett	Engineering/construction	Daily
Operations Manager	Carlos Ramirez	Plant operations	Daily
HSE Manager	Grace Wilson	Safety/environment	Daily/weekly
Procurement Lead	Michael Turner	Equipment/contracts	Weekly
Commissioning Lead	Sarah Patel	Testing/start-up	Twice weekly
Finance Manager	Nina Shah	Cost reporting	Monthly

6. Major Deliverables and Completion

Deliverable	Owner	Baseline	Complete	Current Activity
Crusher upgrade	Engineering	30 Jun 2026	68%	Mechanical installation
Conveyor extension	Engineering	31 Jul 2026	61%	Structural work
Flotation expansion	Process	31 Aug 2026	47%	Equipment installation
Pump station	Water	31 May 2026	72%	Electrical work
Water-recovery system	Water	31 Jul 2026	54%	Civil works
Electrical substation	Electrical	30 Jun 2026	58%	Equipment delivery
Maintenance workshop	Facilities	31 Jul 2026	66%	Building works
Haul-road modifications	Mining	31 May 2026	73%	Earthworks

Deliverable	Owner	Baseline	Complete	Current Activity
Commissioning plan	Commissioning	31 Aug 2026	29%	Drafting
Operator training	Operations	15 Sep 2026	18%	Planning
Production ramp-up	Operations	30 Sep 2026	0%	Not started

7. Schedule Baseline and Current Forecast

Milestone	Baseline	Forecast	Variance	Complete
Engineering freeze	30 Jun 2025	18 Jul 2025	+18 days	100%
Long-lead equipment ordered	30 Sep 2025	22 Nov 2025	+53 days	100%
Civil works complete	31 Mar 2026	30 Jun 2026	+91 days	82%
Crusher installation	30 Jun 2026	18 Sep 2026	+80 days	68%
Conveyor installation	31 Jul 2026	24 Oct 2026	+85 days	61%
Flotation equipment	31 Aug 2026	18 Nov 2026	+79 days	47%
Electrical substation	30 Jun 2026	15 Oct 2026	+107 days	58%
Integrated commissioning	31 Aug 2026	15 Jan 2027	+137 days	0%
Performance testing	15 Sep 2026	15 Feb 2027	+153 days	0%
Production ramp-up	30 Sep 2026	18 Mar 2027	+169 days	0%

8. Detailed Work Plan

ID	Task	Owner	Finish	Complete	Dependency
NR-001	Complete crusher structural works	Civil	28 Aug	74%	Foundation
NR-002	Install crusher drive assembly	Mechanical	18 Sep	42%	NR-001
NR-003	Complete conveyor foundations	Civil	15 Sep	61%	Civil works
NR-004	Install conveyor structure	Mechanical	24 Oct	33%	NR-003
NR-005	Install flotation cells	Process	18 Nov	21%	Equipment delivery
NR-006	Complete pump-station electrical work	Electrical	10 Sep	72%	Equipment
NR-007	Complete substation installation	Electrical	15 Oct	58%	Transformers
NR-008	Complete water-recovery civil works	Water	30 Sep	54%	Earthworks
NR-009	Complete maintenance workshop	Facilities	20 Sep	66%	Building
NR-010	Complete haul-road modification	Mining	28 Aug	73%	Earthworks
NR-011	Develop commissioning procedures	Commissioning	15 Sep	29%	Engineering
NR-012	Mechanical completion inspection	Engineering	15 Dec	0%	All installation
NR-013	Cold commissioning	Commissioning	15 Jan	0%	NR-012
NR-014	Hot commissioning	Operations	15 Feb	0%	NR-013
NR-015	Production ramp-up	Operations	18 Mar	0%	NR-014

9. Budget and Financial Information

Category	Approved	Actual	Committed	Forecast
Civil works	$42.0M	$39.8M	$5.1M	$48.7M
Processing equipment	$68.0M	$51.6M	$18.4M	$82.9M
Electrical infrastructure	$24.0M	$18.7M	$7.6M	$31.8M
Water infrastructure	$18.0M	$12.9M	$4.7M	$22.6M
Contractor services	$31.0M	$26.4M	$8.9M	$39.7M
Engineering/design	$11.0M	$10.2M	$1.0M	$11.8M
HSE/environment	$5.0M	$4.1M	$0.6M	$5.4M
Training/commissioning	$4.0M	$0.8M	$0.9M	$5.7M
Contingency	$17.0M	$2.1M	$0.0M	$12.4M
TOTAL	$220.0M	$166.6M	$47.2M	$260.9M

10. Monthly Cost Trend

Month	Planned	Actual	Variance	Primary Area
Mar 2026	$18.2M	$19.6M	+$1.4M	Civil works
Apr 2026	$19.5M	$21.1M	+$1.6M	Civil/equipment
May 2026	$20.3M	$23.0M	+$2.7M	Equipment
Jun 2026	$21.5M	$24.8M	+$3.3M	Equipment
Jul 2026	$20.8M	$25.4M	+$4.6M	Contractors
Aug MTD	$11.2M	$15.1M	+$3.9M	Electrical/equipment
Sep forecast	$19.8M	$24.5M	+$4.7M	Installation
Oct forecast	$18.5M	$26.2M	+$7.7M	Construction
Nov forecast	$17.4M	$25.8M	+$8.4M	Commissioning

11. Workforce and Resource Position

Work Group	Planned	Available	Availability	Note
Project management	8	6	75%	Two vacancies
Civil construction	140	104	74%	Contractor shortage
Mechanical	96	68	71%	Skilled trades shortage
Electrical	54	38	70%	Specialist shortage
Process engineering	22	17	77%	Shared resources
Commissioning	30	8	27%	Ramp-up not staffed
HSE	18	14	78%	Coverage stretched
Operations	65	49	75%	Existing plant demand
Procurement	16	11	69%	Supplier workload
Maintenance	42	29	69%	Existing plant support

The project is being executed while the existing operation remains active, limiting the number of personnel who can be released for construction, commissioning and training activities.

12. Contractor and Supplier Information

Supplier	Scope	Contract	Finish	Complete	Current Note
OreTech Systems	Crusher equipment	$34M	18 Sep	68%	Late component
ConveyPro	Conveyor system	$22M	24 Oct	61%	Fabrication delay
FloatMet	Flotation equipment	$31M	18 Nov	47%	Two shipments pending
PowerGrid Services	Substation	$19M	15 Oct	58%	Transformer delay
HydroWorks	Water infrastructure	$16M	30 Sep	54%	Civil rework
BuildCore	Civil works	$42M	30 Jun	82%	Variation orders
CommissionPlus	Commissioning	$8M	15 Feb	0%	Mobilization pending

13. Production and Equipment Metrics

Measure	Current	Target / Baseline
Current processing rate	21,700 t/day	22,000 t/day baseline
Target expanded rate	31,000 t/day	31,000 t/day
Crusher installation completion	68%	100%
Conveyor installation completion	61%	100%
Flotation installation completion	47%	100%
Substation installation	58%	100%
Water-recovery construction	54%	100%
Equipment acceptance tests passed	71%	≥95%
Open equipment defects	38	<15
Critical equipment defects	9	0
Rework items, last 60 days	27	<5
Unplanned plant interruptions	14	≤3
Commissioning procedures complete	29%	100% before start-up

14. Safety, Environmental and Compliance Information

The project is subject to site safety procedures, environmental permits, contractor induction requirements, equipment inspection procedures and water-management controls.

During the current reporting period, the site recorded two lost-time incidents, six medical-treatment cases and 19 near-miss reports associated with construction and contractor activities. Corrective-action reviews are ongoing.

Environmental monitoring has identified elevated sediment levels at one drainage-control location following heavy rainfall. Additional containment work has been requested.

Permit documentation for the expanded water-recovery system is still under review. The environmental team is coordinating additional documentation with the relevant authorities.

15. Material, Logistics and Supply Information

The project depends on imported and long-lead mechanical and electrical equipment. Several components are transported through a remote regional corridor with limited heavy-haul capacity.

Material / Equipment	Required	Received	Outstanding	Current Information
Crusher components	100%	72%	28%	One major assembly late
Conveyor steel	100%	64%	36%	Fabrication delay
Flotation cells	100%	47%	53%	Two shipments pending
Transformers	100%	50%	50%	Delivery revised
Large pumps	100%	81%	19%	One unit awaiting inspection
Electrical cabling	100%	76%	24%	Partial deliveries
Structural steel	100%	83%	17%	Additional rework

Several work fronts cannot be closed until equipment and structural materials arrive. Site storage capacity is also constrained by the simultaneous construction of multiple plant areas.

16. Quality and Inspection Records

Quality inspections cover welding, structural steel, equipment foundations, concrete, electrical installation, piping, machine alignment and equipment commissioning.

Inspection Measure	Current	Target
Planned inspections	1,420	1,420
Completed	1,011	100%
Passed first inspection	822	≥95%
Reinspection required	189	<70
Open quality observations	74	<25
Major non-conformances	12	0
Welding rework rate	11.8%	<4%
Concrete rework items	17	<5
Equipment alignment exceptions	23	<5

17. Commissioning and Readiness Information

Integrated commissioning has not started. Mechanical completion is dependent on several construction work fronts and equipment deliveries.

The commissioning team currently has eight available specialists against a planned group of 30. Mobilization of additional commissioning personnel is scheduled but has not yet been completed.

Operating procedures are being drafted while installation continues. The current commissioning package covers approximately 29% of planned procedures.

The existing processing plant must remain operational during construction, which limits available shutdown windows for tie-ins and testing.

18. Stakeholder and Communication Information

Weekly project meetings have averaged 78% attendance over the last eight meetings. Contractor attendance has varied between 62% and 84%. Operations representatives frequently participate remotely because of active plant duties.

The executive sponsor attended two of the last five steering meetings. The site general manager has requested additional weekly reporting on schedule, contractor productivity and material deliveries.

Finance has requested a revised forecast because the latest estimate exceeds the approved project budget. Procurement has requested decisions on several supplier variations.

19. Recent Meeting Records

05 August — Construction Review: Civil teams reported additional rework on foundations and drainage structures.

08 August — Equipment Review: The crusher supplier reported a delayed major assembly. Revised delivery timing was not yet confirmed.

11 August — Safety Review: Contractor safety performance was reviewed following a lost-time incident and several near-miss reports.

13 August — Finance Review: The latest cost forecast exceeded the approved budget and additional contractor variations were discussed.

15 August — Operations Review: Plant shutdown windows for tie-ins were reduced because of production requirements.

17 August — Commissioning Review: Commissioning staffing was reviewed; only a small portion of planned specialists were currently available.

18 August — Steering Review: Leadership requested an integrated recovery schedule covering equipment, construction, commissioning and production ramp-up.

20. Pending Actions and Upcoming Dates

Task	Owner	Due	Progress
Confirm crusher component delivery	Procurement	25 Aug	Pending
Complete crusher structural works	Civil	28 Aug	74%
Complete conveyor foundations	Civil	15 Sep	61%
Complete pump-station electrical work	Electrical	10 Sep	72%
Resolve transformer delivery schedule	Procurement	29 Aug	Pending
Complete water-recovery civil works	Water	30 Sep	54%
Complete commissioning procedures	Commissioning	15 Sep	29%
Mobilize commissioning specialists	Operations	05 Sep	27%
Mechanical completion inspection	Engineering	15 Dec	0%
Cold commissioning	Commissioning	15 Jan	0%
Hot commissioning	Operations	15 Feb	0%
Production ramp-up	Operations	18 Mar	0%

21. Change Requests and Variations

CR	Request	Requested By	Cost / Effort	Status
NR-CR-18	Additional foundation reinforcement	Engineering	$2.4M	Approved
NR-CR-19	Drainage containment modification	HSE	$1.1M	Under review
NR-CR-20	Conveyor structural redesign	Engineering	$3.8M	Under review
NR-CR-21	Additional contractor shifts	Construction	$4.6M	Approved
NR-CR-22	Extended equipment storage	Logistics	$0.9M	Approved
NR-CR-23	Additional commissioning support	Operations	$2.1M	Pending
NR-CR-24	Temporary power arrangement	Electrical	$1.4M	Under review

22. Constraints and Assumptions

Construction is taking place beside an operating mine and processing plant. Access to work areas is controlled by production schedules and safety requirements.

Heavy equipment deliveries are dependent on a limited regional transportation corridor. Weather conditions can affect earthworks and road access.

Commissioning cannot proceed until sufficient mechanical completion is achieved and specialist personnel are available.

Several supplier delivery dates remain subject to confirmation. Existing production requirements limit available plant shutdown windows.

The current cost forecast assumes additional contractor productivity measures and successful completion of outstanding equipment deliveries.

23. Raw Project Summary

Measure	Current Value
Approved budget	$220.0M
Actual expenditure	$166.6M
Committed expenditure	$47.2M
Current forecast	$260.9M
Original completion	30 Sep 2026
Current forecast completion	18 Mar 2027
Planned workforce	453
Available workforce	324
Open equipment defects	38
Critical equipment defects	9
Equipment acceptance pass rate	71%
Quality first-pass rate	81.3%
Major non-conformances	12
Lost-time incidents	2

Measure	Current Value
Near-miss reports	19
Unplanned plant interruptions	14
Commissioning completion	0%
Commissioning procedure completion	29%
Material outstanding across major equipment	19–53% by category
Current forecast budget variance	+$40.9M
Current schedule variance	+169 days

24. Document Boundary

This is intentionally a **raw project information source**. It does not explicitly state a risk level, risk probability, health score, blocker classification, future outcome or recommendation. Those outputs are expected from the AI Project Intelligence & Risk Advisor. The organization, people, contractors, financial figures and operational data are fictional and created solely for model testing.